FACULTY OF COMPUTING

SEMESTER 2

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SECI 1143 - PROBABILITY AND STATISTICAL DATA ANALYSIS

SECTION 02

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CASE STUDY: HEART DISEASE PREDICTION

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1. **Introduction**

Heart disease, also known as cardiovascular disease, encompasses a variety of conditions and disorders that impact the heart and circulatory system. It stands as a leading cause of disability globally. Given the heart's critical role in maintaining overall health, diseases affecting it can have widespread consequences on other organs and bodily functions. Our interest in this data stems from a desire to understand the differences in patient attributes and their influence on disease progression. We aim to uncover the relationship between various health factors and their potential role in exacerbating the diseases.

2. **Dataset**

2.1 Data Description

The population of the study in this data is 207 male and 96 female with the youngest age is 29 years old and the oldest age is 77 years old. From the dataset, the data description of the study is represented by the variable in *Table 1* below.

Variables	**Description**	**Type of Variable**	**Measurement Level**
age	Age of the population	Quantitative	Ratio
sex	Gender of the population 0: Female 1: Male	Quantitative	Nominal
cp	Chest pain type 0: Typical angina 1: Atypical angina 2: Non-anginal pain 3: Asymptomatic	Quantitative	Nominal
trestbps	Resting blood pressure (in mm Hg on admission to the hospital)	Quantitative	Interval
chol	serum cholesterol in mg/dl	Quantitative	Ratio
fbs	Fasting blood sugar (>120 mg/dl) 0: False 1: True	Quantitative	Nominal
restecg	Resting electrocardiographic results	Quantitative	Nominal

	0: Normal 1: Having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) 2: Showing probable or definite left ventricular hypertrophy by Estes' criteria		
thalach	Maximum heart rate achieved	Quantitative	Ratio
exang	Exercise induced angina 0: No 1: Yes	Quantitative	Nominal
oldpeak	ST depression induced by exercise relative to rest	Quantitative	Ratio
slope	Slope of the peak exercise ST segment 0: Upsloping 1: Flat 2: Downsloping	Quantitative	Nominal
ca	Number of major vessels (0-3) coloured by fluoroscopy	Quantitative	Ratio
thal	Thalassemia 1: Fixed defect 2: Normal 3: Reversible defect	Quantitative	Nominal
target	The label 0: No disease 1: Disease	Qualitative	Nominal

Table 1: Data Description

2.2 Data Pre-Processing

From the dataset, in column thal in data #50 and 283, the data is NaNs (Not a Number) which represent undefined values. While in column ca in data #94, 160, 165, 166 and 252 also have NaNs (Not a Number) which represent undefined values. Therefore, the 7 faulty data of the original dataset will be dropped. This resulted in a new sample size in which changes from 303 population to 296.

2.3 Statistical Test Analysis

The test analysis of the objective and the expected outcome of the selected variable will be explained in *Table 2* below.

Selected Variable	**Objective**	**Test Analysis and Expected Outcome**
age, target (heart disease status)	To determine whether there is significant difference in the mean age between individuals with heart disease and those without heart disease	**Test Analysis:** Hypothesis Testing 2-Sample **Expected Outcome:** There is no significant difference in mean age between individuals with and without heart disease based on the data analyzed
age, chol (serum cholesterol)	To determine the relationship between age and serum cholesterol to identify how age and serum cholesterol strongly related to each other as age and serum increases	**Test Analysis:** Correlation Analysis **Expected Outcome:** A positive correlation is found where as the age increases, the serum cholesterol also increases.
age and trestbps (Resting blood pressure) Independent variable: Age Dependant variable: Resting blood pressure	To determine the relationship between age and resting blood pressure and if any change in age will have any impact towards resting blood pressure.	**Test Analysis:** Regression Test **Expected Outcome:** The relationship between age and resting blood pressure is a positive linear relationship.
sex and exang (Exercise Induced Angina)	To determine whether the variables gender and exercise induced angina are related or independent.	**Test Analysis:** Chi Square test of independence **Expected Outcome:** There is a relationship between the variables sex and exercise induced angina

Table 2: Statistical Data Analysis

3. Data Analysis

Four tests are carried out from the chosen dataset which are Hypothesis testing 2-sample, Correlation test, Regression test and Chi-square test of independence. Below is the analysis of each test.

3.1 Hypothesis Testing 2-sample

In this analysis, we use variable age and target where we will test whether the mean target of people with no heart disease is different from the mean age of people with heart disease. From the data, frequency(n), mean($\bar{x}$), standard deviation(s) are calculated.

```
              Group Frequency Mean_Age    SD_Age
0 No Heart Disease        137 56.63504 7.981482
1    Heart Disease        164 52.49390 9.579821
```

Figure 1: Calculation of Data Collected in RStudio

From the Figure 1, we know that:

Heart Disease	**Frequency(n)**	**Mean Age($\bar{x}$)**	**Standard Deviation of Age(s)**
No (0)	137	56.64	7.98
Yes(1)	164	52.49	9.58

Table 3: Summarized Data for Hypothesis Testing from RStudio

Hypothesis Statement:

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

Where the μ_1 is the mean age of people with no heart disease and μ_2 means the mean age of people who have heart disease.

Given the confidence level is 95%, $\alpha = 0.05$.

By using the RStudio, the test statistic, $t_0 = 4.100$

t0	4.10021277590683

Figure 2: Test Statistic by RStudio

By using the RStudio, the degree of freedom, $v = 298.999 \approx 298$

v	298.998500915352

Figure 3: Degree of Freedom by RStudio

The critical value from RStudio = -1.97

t.alpha	-1.96795650649682

Figure 4: Critical Value by RStudio

Critical value, $t_{-0.025,298}$ = -1.97 and $t_{0.025,298}$ = 1.97
Therefore, using $\alpha = 0.05$, we reject H_0 if $t_0 < t_{0.025,298}$ or $t_0 > t_{0.025,298}$

Conclusion:
Since t_0 = 4.100 > critical value which is $t_{0.025,298}$ = 1.97, we reject the null hypothesis, H_0. Hence, there is sufficient evidence to conclude that the mean age of people with no heart disease is different from the mean age of people who have heart disease.

```
        Welch Two Sample t-test

data:  age by target
t = 4.0912, df = 299, p-value = 5.525e-05
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 2.149161 6.133107
sample estimates:
mean in group 0 mean in group 1
       56.63504        52.49390
```

Figure 5: Two Sample t-test Output using RStudio

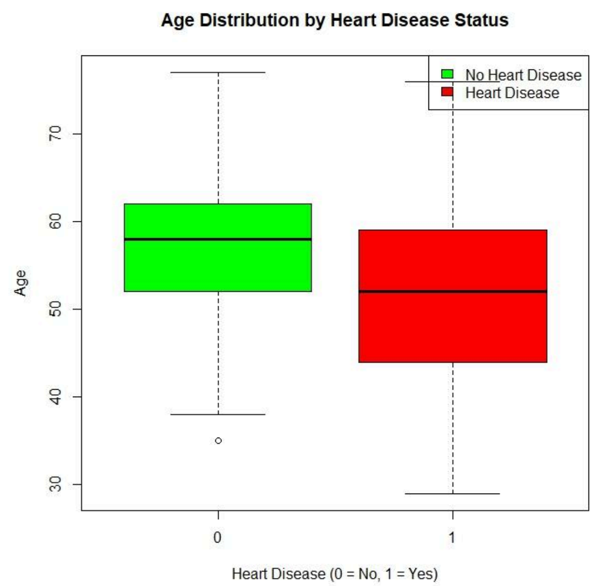

Figure 6: Data Visualized Using Boxplot

3.2 Correlation Test

In this analysis, we will use age and chol variables to determine how strongly related are between age and chol as the age increases. As the measurement level of variable age and chol are both ratio-type, we will use the Pearson's Product-Moment Correlation Coefficient for the correlation coefficient, with 95% confidence level, we also will determine the linear relationship between the two variables.

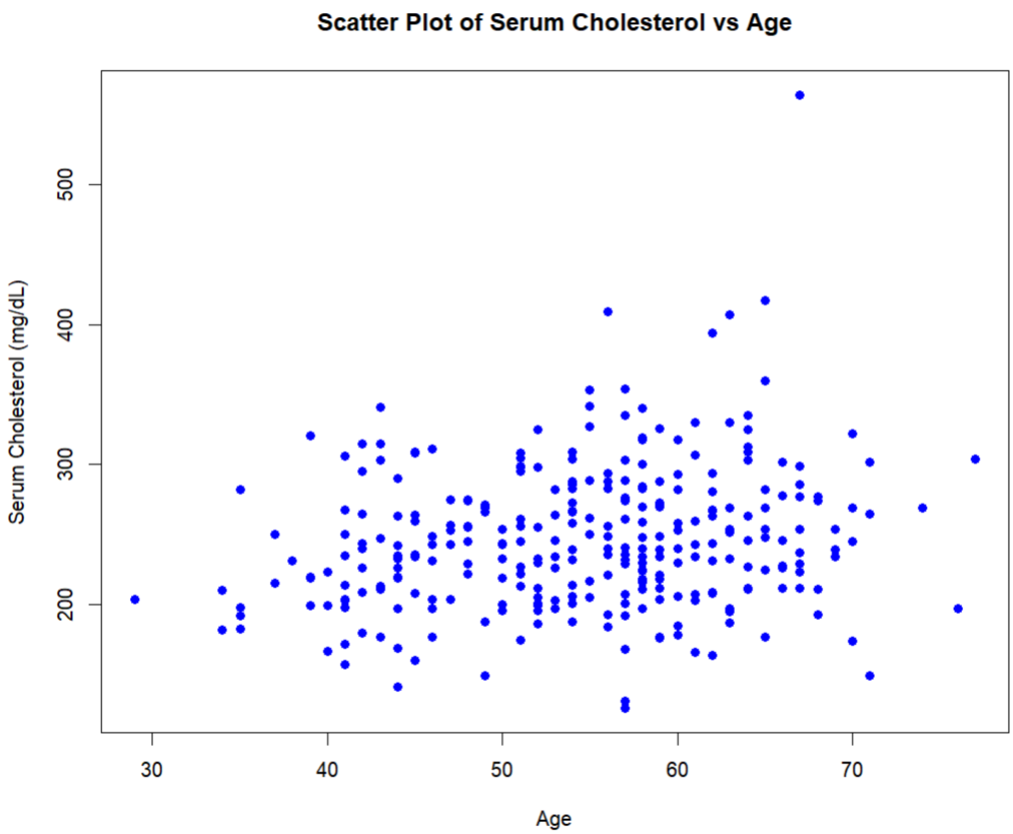

Figure 7 : Scatter Plot of Serum Cholesterol and Age using RStudio

According to the scatter plot in Figure 7 below, the distribution of the plot shows that the data point is widely scattered across the age range from 30 to 70 years old as the youngest age is 29 years old while the oldest age is 77 years old and serum cholesterol levels is from around 150 mg/dL to over 500 mg/dL. The pattern of the scatter plot does not show a clear linear pattern where the points are scattered without a discernible trend.

Calculate the sample correlation using Pearson's method:

```
> # Calculate and print the correlation
> r <- cor(y, x)
> cat("Sample Correlation Coefficient: ", r, "\n")
Sample Correlation Coefficient:  0.2009196
```

Figure 8 :Correlation Coefficient using RStudio

Based on the Figure 8 shown, the sample correlation coefficient is 0.2009196 which is close to 0. The results indicate that there is a relatively weak positive linear relationship between variable age and chol, which shows that as the age increases, the cholesterol also increases but is not strongly related.

Significance Test For Correlation:

To determine whether the variable age and chol is statistically significant not, we will be using test statistics to determine.

Hypothesis Statement:

$H_0 : \rho = 0$ (No Linear Correlation)

$H_1 : \rho \neq 0$ (Have Linear Correlation)

Test Statistic:

```
> # Calculate the test statistic
> n <- length(x)
> t_stat <- r * sqrt((n - 2) / (1 - r^2))
> cat("Test Statistic (t): ", t_stat, "\n")
Test Statistic (t):  3.516768
```

Figure 9 : Test Statistic using RStudio

Figure 9 below shows that, the test statistic, t is $t = 3.516768$.

Critical value:

The value of the critical value is find using $\alpha = 0.05$, and the degree of freedom is $df = 294$.

Since the test is two-tailed test, we will have 2 critical values where:

- Lower tail = -1.97
- Upper tail = 1.97

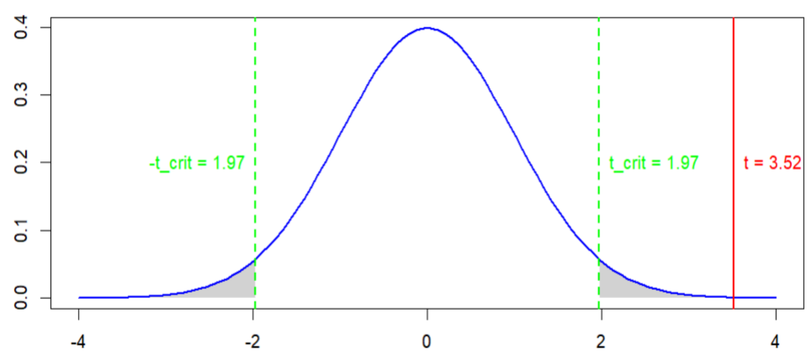

Figure 10: Normal Distribution Graph For Correlation

Conclusion:

```
> # Print a summary of the correlation test
> cor_test <- cor.test(x, y)
> print(cor_test)

	Pearson's product-moment correlation

data:  x and y
t = 3.5168, df = 294, p-value = 0.0005058
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.08895261 0.30787209
sample estimates:
      cor
0.2009196
```

Figure 11 : Correlation Result using RStudio

Since $t = 3.516768$, higher than the upper tail $t_{0.025/294} = 1.97$, we will reject the null hypothesis, H_0. To conclude, the evidence provided is sufficient to conclude that there is a linear relationship between the variable age and chol at $\alpha = 0.05$. Moreover, since the correlation coefficient is 0.2009196 which is close to 0, indicates a weak positive correlation between age and chol, whereas as the age increases, the chol also increases. Since the relationship shown is weak, it indicates that the other factors also play a role in influencing serum cholesterol where it is not done by looking at age only.

3.3 Regression Test

In this test, we take two variables from the dataset to test:

Dependent variable (y) = Resting blood pressure

Independent variable (x) = Age

We use RStudio to find the property for regression analysis. Below are the findings from RStudio code:

```
> summary(model)

Call:
lm(formula = y ~ x)

Residuals:
    Min      1Q  Median      3Q     Max 
-38.439 -11.499  -1.044  10.192  67.495 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 102.2961     5.8906  17.366  < 2e-16 ***
x             0.5394     0.1069   5.048 7.76e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.87 on 301 degrees of freedom
Multiple R-squared:  0.07804,	Adjusted R-squared:  0.07497 
F-statistic: 25.48 on 1 and 301 DF,  p-value: 7.762e-07
```

Figure 12: Rstudio regression summary

Estimated Regression Model:

$$\hat{y} = 102.2961 + 0.5394x$$

Here, none of the people has age is equal to zero, so b_0 = 102.2961 just indicates that, for people within the range of sizes observed, 102.2961 is the portion of the resting blood pressure not explained by age. Meanwhile, for b_1 = 0.5394 tells us that the average value of resting blood pressure increases by 0.5394 on average, for each additional year of age.

Multiple R-squared = 0.07804

Only 7.8% of the variation in resting blood pressure is explained by variation in age.

$$S_\varepsilon = 16.87$$

$$S_{b1} = 0.1069$$

Variation of observed y values from the regression line is large. The variation in the slope of the regression line from different possible samples is also large. This two can be observed from the graph in Figure 3:

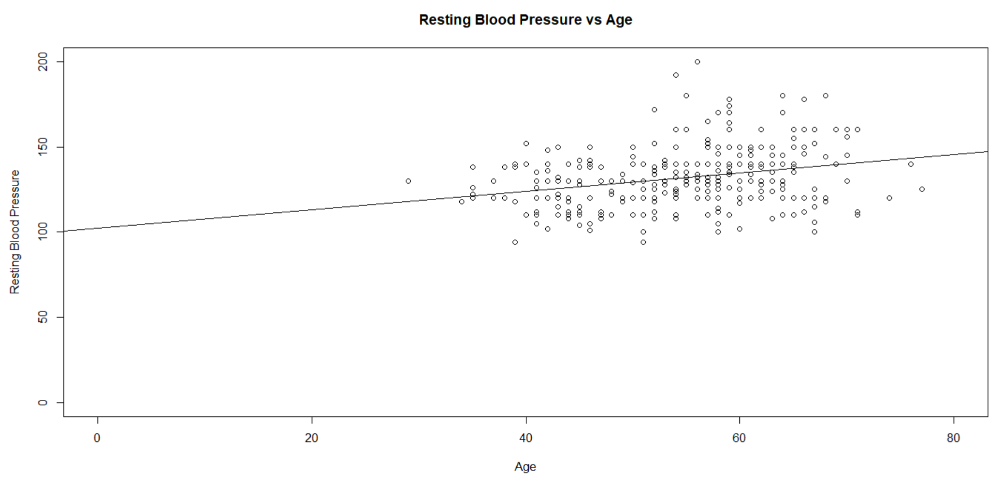

Figure 13: Scatter Plot Graph of Resting Blood Pressure vs Age

To find if there is a linear relationship between x and y:

H_0: $\beta_1 = 0$ (no linear relationship)

H_1: $\beta_1 \neq 0$ (linear relationship does exist)

From RStudio calculation, we got that,

$t = 5.048$

d.f. = 296-2 = 294; $\alpha = 0.05$

$t_{\alpha/2} = 1.960$ (refer to table)

$1.960 < 5.048$

Decision: Reject H_0

Conclusion: There is sufficient evidence to reject the claim that there is no linear relationship. Age does affect resting blood pressure.

3.4 Chi Square test of independence

In this analysis, we use the variables sex and exang to assess the independence of gender and exercise-induced angina using a Two Way Contingency Table at a 95% confidence level. As a result, we employ the Chi-Square Test of Independence in conjunction with the two-way contingency table.

```
> table(sex, exang)
   exang
sex   0   1
  0  73  22
  1 130  76
```

Figure 14: Two Way Contingency Table using RStudio

Hypothesis statement:

H_0: No relationship between variables.

H_1: Variables have relationships.

Critical value:

By using RStudio, we obtain the Critical value, $x^2 = 3.841$, with α=0.05 and degree of freedom, df=(2-1)(2-1)=1.

```
> df_1 = 1
> alpha = 0.05
> t_alpha = qchisq(alpha, df_1, lower.tail=FALSE)
> t_alpha
[1] 3.841459
```

Figure 15: Critical Value using RStudio

Expected counts:

Sex	**Exercise-Induced Angina**				**Total**
	0		**1**		
	obs.	**exp.**	**obs**	**exp.**	
0	73	64.070	22	30.930	95
1	130	138.930	76	67.070	206
Total	203	203.000	98	98.000	301

Table 4: Expected Counts Calculated Manually

```
> result$expected
   exang
sex        0        1
  0  64.06977 30.93023
  1 138.93023 67.06977
```

Figure 16:Expexted Counts using RStudio

Test statistic value:

Cell, ij	Observed Count, o_{ij}	Expected Count, e_{ij}	$\frac{(o_{ij}-e_{ij})^2}{e_{ij}}$
1,1	73	64.070	1.245
1,2	22	30.930	2.578
2,1	130	138.930	0.574
2,2	76	67.070	1.189
		x^2	5.586

Table 5: Test Statistic Calculated Manually

When we calculate test statistic manually, we get test statistic, x^2= 5.586.

```
        Pearson's Chi-squared test with Yates' continuity correction

data:  a
X-squared = 4.9781, df = 1, p-value = 0.02567
```

Figure 17 : Test Statistic using RStudio

When we calculate test statistic using RStudio, we get test statistic,x^2= 4.978, with the p-value = 0.026.

Conclusion:

Since the test statistic value (x^2= 4.978) > critical value ($x^2_{k=1,\, \alpha=0.05}$ = 3.841), it does fall within the critical region. Thus, we reject the null hypothesis, H_0. Therefore, there is sufficient evidence to conclude that there is a relationship between the variables sex and exang, at α = 0.05.

4. Conclusion

According to the hypothesis test, there is a significant difference in the mean ages of those with and without heart disease. The calculated t-value was 4.100, which exceeds the critical value of 1.97 at a 95% confidence level. As a result, we reject the null hypothesis, H_0 and come to the conclusion that there is a difference in the mean age between those with and without heart disease. Furthermore, based on the boxplot, we can see it appears that the individuals with heart disease tend to be older.

The outcomes of the correlation test between age and serum cholesterol (chol) shows a weak positive linear relationship, with a correlation coefficient of 0.201, suggesting that cholesterol levels increase along with age but not significantly. A statistically significant linear relationship between age and cholesterol is verified by the significance test, which yields a test statistic of t=3.517, exceeding the critical value of 1.97 at a 95% confidence level and leads to the rejection of the null hypothesis, H_0.

Next, for the regression test, we found that the scatter plot graph shows a positive linear association between age and resting blood pressure. The regression model, $\hat{y} = 102.2961 + 0.5394x$ indicates that as age increases, the resting blood pressure will rise by approximately 0.5394 mm Hg per year. The regression coefficient that was obtained was 5.048, which is higher than the critical value. As a result, we reject the null hypothesis and conclude that age has a significant effect on resting blood pressure.

The chi-square test of independence between sex and exercise-induced angina (exang) reveals a significant relationship between the two variables. As the test statistic of 4.978 surpasses the 95% confidence level critical value of 3.841, the null hypothesis, H_0 is rejected. This result indicates that gender does affect the occurrence of exercise-induced angina, demonstrating a dependence between these variables

With RStudio, we were able to carry out a number of tests, including the 2 sample hypothesis testing, correlation test, regression test and chi-square test of independence. The result emphasized the importance of features such as age, sex, cholesterol levels and resting blood pressure in predicting heart disease.

5. Appendix

Data Resource Link

Desalegngeb. (2023, April 30). Heart Disease Predictions. Retrieved from: **Heart Disease Predictions (kaggle.com)**

Raw Dataset

HEART DATA.xlsx

Processed Dataset

DATA PROJECT 2.xlsx

Reflection

REFLECTION

Presentation Video

YouTube link